CSE 1321L: Programming and Problem Solving I Lab

Assignment 3 – 100 points
Solving Problems

What students will learn:

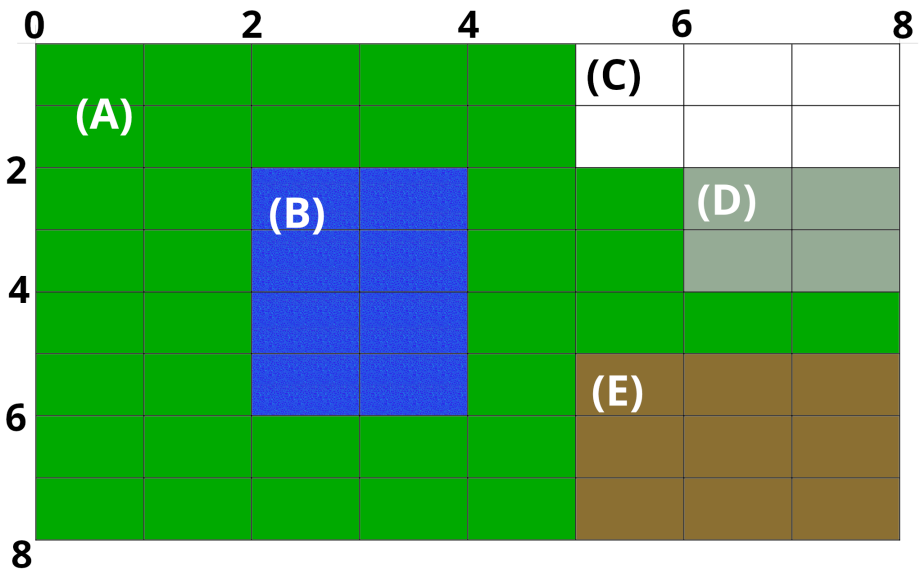
1) Problem solving
2) Structure programs to include conditional logic
3) Write code that includes if/else statements
4) Design programs that leverage switch/case logic
5) Solve problems of increasing complexity

Overview: For this assignment, you're going to practice making decisions in your code. It's because of those decisions that your program can behave differently depending on the values in your variables; these decisions are called **conditional statements**. In practical terms, this means you're going to expand on the concepts from assignment 1 and 2, but also include things like IF, ELSE IF, ELSE and SWITCH statements. You'll also work with something called **compound conditionals**, which is just a fancy way of saying that you're going to use AND and OR operators (&& and ||). Again, start early, practice, and ask a lot of questions.

Follow the same conventions for class names and file names for your source code. For the Java folks, remove the "package" statement if you have one. Make sure to follow the FYE Submission Guidelines. Finally, we don't mean to lecture, but we want to remind you not to cheat. This is the core of what a lot of you will be doing for a living, so master it now.

Assignment3A:

Cretaceous Park Map: Noted business man John Hammond is reviving his dinosaur-themed amusement park after a disastrous first attempt in 1993. He has hired you to develop a Geographic Information System (GIS) to quickly identify where someone is in the park based on their coordinates. A grid-based map of the park is shown below:



Legend

(A)	Triceratops Grasslands
(B)	T. Rex Lake
(C)	Pterodactyl Aviary
(D)	Visitor's Center
(E)	Deinonychus Pen

You will prompt the user to enter in their X and Y coordinates, and then tell them where they are in the park. You should also warn them if they have strayed outside the park boundaries.

You must call the class **Assignment3A** and put it in a file called "Assignment3A.java", "Assignment3A.cs", or "Assignment3B.cpp"
You will need to use nested or compound IF statements to implement this program.

Sample Output #1:

```
[Cretaceous Park GIS]
X Coordinate: 10
Y Coordinate: -20

WARNING: You have left the boundaries of the park.
Please return immediately.
```

Sample Output #2:

```
[Cretaceous Park GIS]
X Coordinate: 6.1
Y Coordinate: 2.2

You are at the Visitor's Center
```

Sample Output #3:

```
[Cretaceous Park GIS]
X Coordinate: 2
Y Coordinate: 2

You are at the T. Rex Lake
```

Assignment3B:

Image Color Depth: In Assignment 1, we learned that each pixel of a computer image is represented with three values, RGB. Traditionally one byte is used for each color, meaning that each pixel takes up 3 bytes of space. These are known as 8 bits per color channel encoding, or 8 BPC for short. More recently, encodings with larger ranges of color have become popular with artists and photographers. Programs like Photoshop now support 16 BPC and 32 BPC encoding.

For this assignment, you will try to determine the encoding of an RGB image based on its width, height, and file size. After the user enters this information, you will first check if the information is valid. If not, you will notify the user and skip to the end of the program.
(*Hint: You will have to check multiple potential issues with your IF statements*)

If the detail is valid, you will then calculate the BPC encoding. Based on the value, you will use a Switch statement to print out if the image is 8, 16, or 32 BPC. If it is outside the range, inform the user that your computer does not know how to read this encoding.

You must call the class **Assignment3B** and put it in a file called "Assignment3B.java", "Assignment3B.cs", or "Assignment3B.cpp". Make sure to use **Switch statements** to complete this assignment – you will lose points if you do not.

Sample Output #1:

```
[Image Encoding Checker]
What is the image width? 10
What is the image height? -20
What is the file size (in bytes)? -200

The information is invalid - please re-enter it.
```

Sample Output #2:

```
[Image Encoding Checker]
What is the image width? 10
What is the image height? 20
What is the file size (in bytes)? 202

The information is invalid - please re-enter it.
```

Sample Output #3:

```
[Image Encoding Checker]
What is the image width? 50
What is the image height? 50
What is the file size (in bytes)? 7500

The RGB image is encoded with 8 bits per channel.
```

Sample Output #4:

```
[Image Encoding Checker]
What is the image width? 100
What is the image height? 100
What is the file size (in bytes)? 60000

The RGB image is encoded with 16 bits per channel.
```

Assignment3C:

Decision Trees: Many videos games and interactive stories use “decision trees” to allow for different dialogue based on the player’s choices. For this assignment, you will create a (basic) dialogue system using **NESTED IF** statements. The diagram on the next page shows the different choices you can make, and the appropriate responses.

Note: The following diagram is a simplified form of UML – Unified Modeling Language. It is one way to display an algorithm so that people can understand it and implement it in code. For our purposes, the boxes show the printed prompts and responses. The diamonds indicate decisions after each prompt. For example, if I respond “No, I just turned off the light.” to the first prompt, my program should print the response shown at the end of the “No” line on the diagram.

You must call the class **Assignment3C** and put it in a file called “Assignment3C.java”, “Assignment3C.cs”, or “Assignment3C.cpp”

Make sure to use **nested IF statements** to complete this assignment. Your program should produce the following output based on different user input (Don’t worry about invalid input).

Sample Output #1:

```
[Cretaceous Park Dialog]
Oh no, the power has gone out at Cretaceous Park!
1) What does that mean?
2) No, I just turned the light off.
No, I just turned off the light

Oh thank goodness!
```

Sample Output #2:

```
[Cretaceous Park Dialog]
Oh no, the power has gone out at Cretaceous Park!
1) What does that mean?
2) No, I just turned the light off.
What does that mean?

The dinosaurs will get loose... again...
1) Life finds a way...
2) What do we do?
Life finds a way...

That’s not helpful right now!
We have to evacuate the park!
```

Sample Output #3:

```
[Cretaceous Park Dialog]
Oh no, the power has gone out at Cretaceous Park!
1) What does that mean?
2) No, I just turned the light off.
What does that mean?

The dinosaurs will get loose... again...
1) Life finds a way...
2) What do we do?
What do we do?

We have to evacuate the park!
```

Submission:

1. You will submit 3 separate files containing source code – one for each of the assignments above. Make sure the files are named correctly and to include the header with your name and course section listed.
2. Upload all 3 files (simultaneously) to the assignment submission folder in Gradescope.
3. We’ll continue to work with you on this assignment if something messes up, so long as you submit by the due date. However, start early because we work during the weekday.

APPENDIX – Examples of testing string equality in Java, C# and C++

```java
//========= Java ==========
import java.util.*;
class Assignment3 {
  public static void main(String[] args) {
    Scanner scan = new Scanner (System.in);
    String userInput;
    System.out.println ("Enter something:");
    userInput = scan.nextLine();
    if (userInput.equals("Bob")) {
      System.out.println ("You typed Bob");
    }
    else {
      System.out.println ("You didn't type Bob");
    }
  }
}
```

```csharp
//========= C# ============
using System;
class Assignment3 {
  public static void Main (string[] args) {
    string userInput;
    Console.WriteLine ("Enter something:");
    userInput = Console.ReadLine();
    if (userInput.Equals("Bob")) {
      Console.WriteLine ("You typed Bob");
    }
    else {
      Console.WriteLine ("You didn't type Bob");
    }

  }
}
```

```cpp
//========= C++ ============
#include <iostream>
#include <cstdlib>
#include <string>

using namespace std;

int main() {
  string userInput;
```

```
  cout << "Enter something:" << endl;
  getline (cin, userInput);
  if (userInput.compare("Bob") == 0) {
    cout << "You typed Bob" << endl;
  }
  else {
    cout << "You didn't type Bob" << endl;
  }
}
```